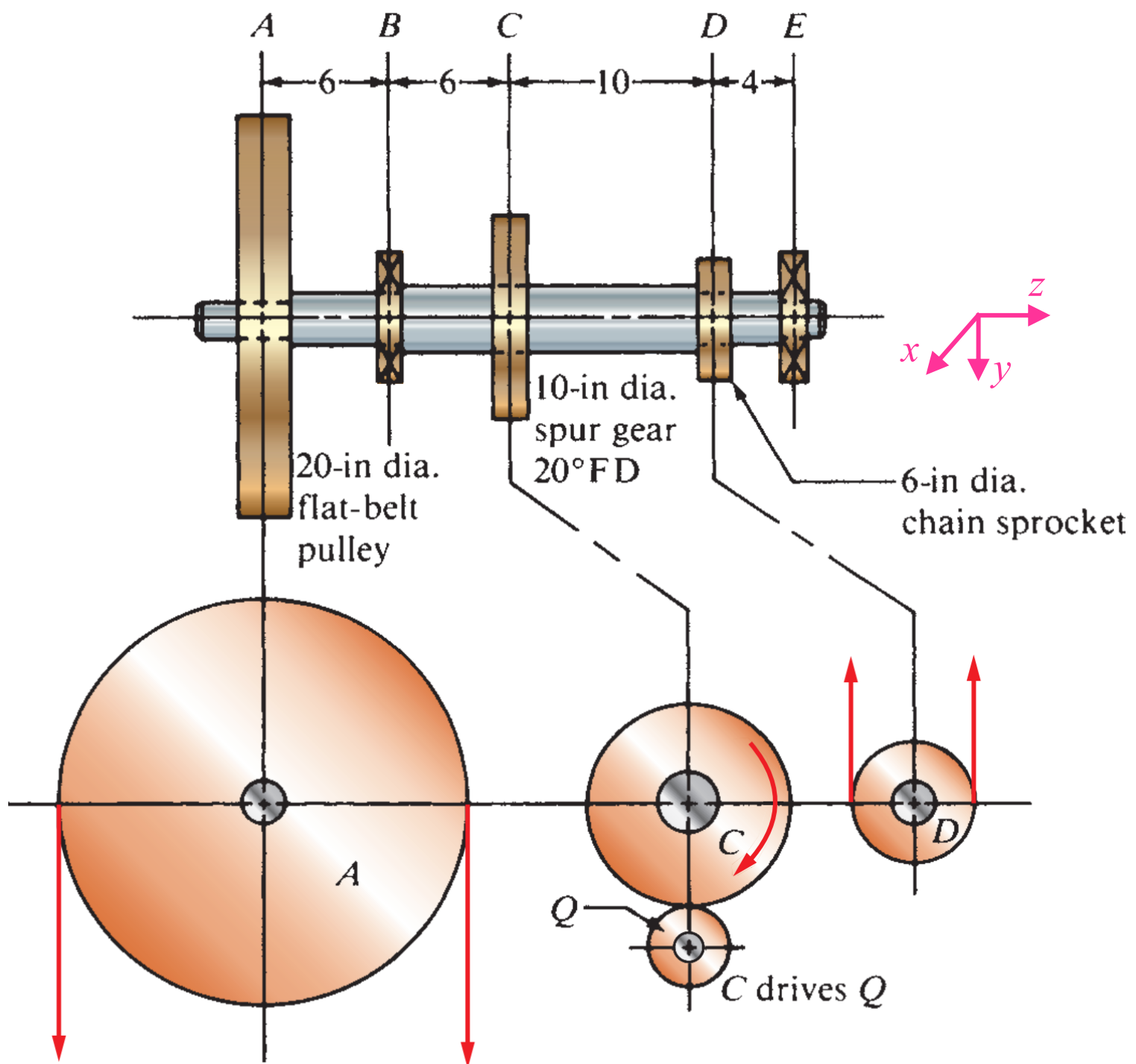


1. The shaft shown in the figure (see next page) rotates at 200 rpm. The pulley *A* receives 20 hp from below. The gear *C* delivers 12 hp to the mating gear below it, and the chain sprocket *D* delivers the rest 8 hp to a shaft above. The shaft material is SAE 1117 cold-drawn steel. Take $C_s = 0.80$, $C_R = 0.81$, design factor $N = 2$, stress concentration factor $K_t = 3$. **(20' in total)**

- 1) Determine the torques on pulley *A*, gear *C*, and sprocket *D*. Also determine the torque distribution in *A-C*, *C-D*, and *D-E* region.
- 2) Plot the shear and bending moment diagrams of the entire shaft in the horizontal and vertical planes.
- 3) Determine the total bending moment at point *B*, *C*, and *D*.
- 4) Determine the minimum acceptable diameters for the shaft at point *C* **(do design to the left and to the right separately)**.



Solutions to Assignment #1:

Step 1: Analyze the torque distribution in the shaft.

The total 20 hp power is split from *A* to the gear *C* by 12 hp and then to the sprocket at *D* by the rest 8 hp.

Torque on pulley *A*:

$$T_A = \frac{63000(20 \text{ hp})}{200 \text{ rpm}} = \boxed{6300 \text{ lb} \cdot \text{in}}$$

Torque on gear *C*:

$$T_C = \frac{63000(12 \text{ hp})}{200 \text{ rpm}} = \boxed{3780 \text{ lb} \cdot \text{in}}$$

Torque on sprocket *D*:

$$T_D = \frac{63000(8 \text{ hp})}{200 \text{ rpm}} = \boxed{2520 \text{ lb} \cdot \text{in}}$$

Torque distribution in the shaft:

$$T_{A-C} = 6300 \text{ lb} \cdot \text{in}$$

$$T_{C-D} = 2520 \text{ lb} \cdot \text{in}$$

$$T_{D-E} = 0 \text{ lb} \cdot \text{in}$$

Solutions to Assignment #1, cont'd:

Step 2: Compute forces that act in the vertical and horizontal planes.

(1) Forces on pulley *A*:

$$\text{Net driving force: } F_N = F_1 - F_2 = \frac{T_A}{\frac{D_A}{2}} = \frac{6300}{\frac{20}{2}} = 630 \text{ lb}$$

$$\text{Bending force at } A: F_A = 2.0F_N = 2.0(630) = 1260 \text{ lb}$$

(Note: for flat-belt pulley, a factor of 2.0 should be used)

The components of the bending force at point *A* are:

$$F_{Ax} = 0 \text{ lb}$$

$$F_{Ay} = 1260 \text{ lb}$$

(2) Forces on gear *C*:

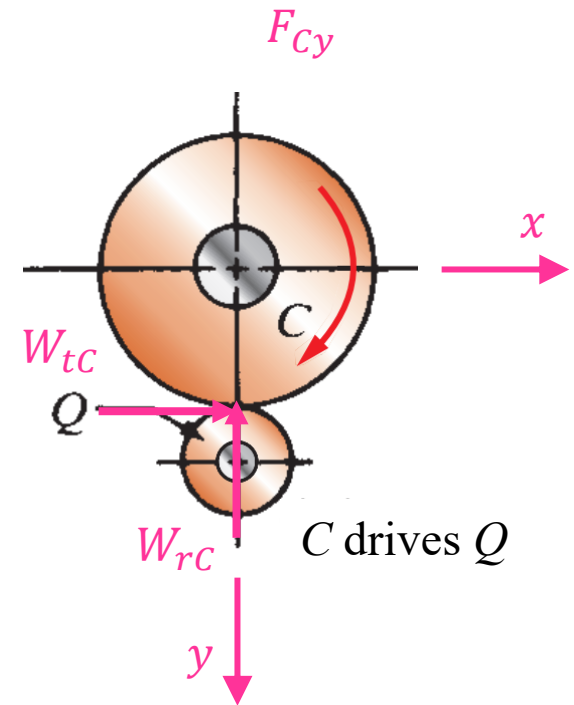
$$W_{tC} = \frac{T_C}{\frac{D_C}{2}} = \frac{3780}{10/2} = 756 \text{ lb}$$

$$W_{rC} = W_{tC} \tan 20^\circ = 756(\tan 20^\circ) = 275 \text{ lb}$$

The components of the bending force at point *C* are:

$$F_{Cx} = W_{tC} = 756 \text{ lb}$$

$$F_{Cy} = -W_{rC} = -275 \text{ lb} \quad (\text{upward or negative } y\text{-direction})$$



Solutions to Assignment #1, cont'd:

(3) Forces on sprocket D :

$$F_D = \frac{T_D}{\frac{D_D}{2}} = \frac{2520}{6/2} = 840 \text{ lb}$$

The components of the bending force at point D are:

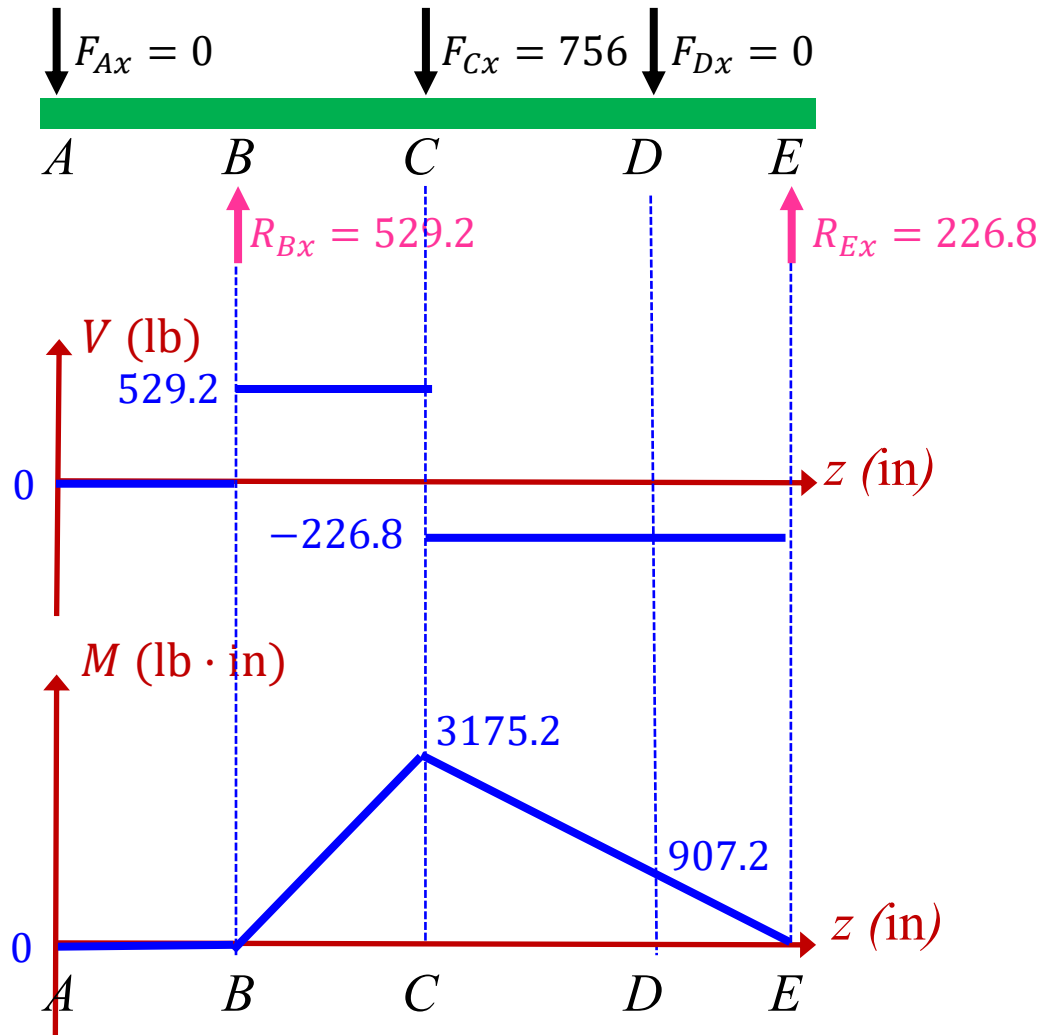
$$F_{Dx} = 0 \text{ lb}$$

$$F_{Dy} = -F_D = -840 \text{ lb} \quad (\text{upward or negative y-direction})$$

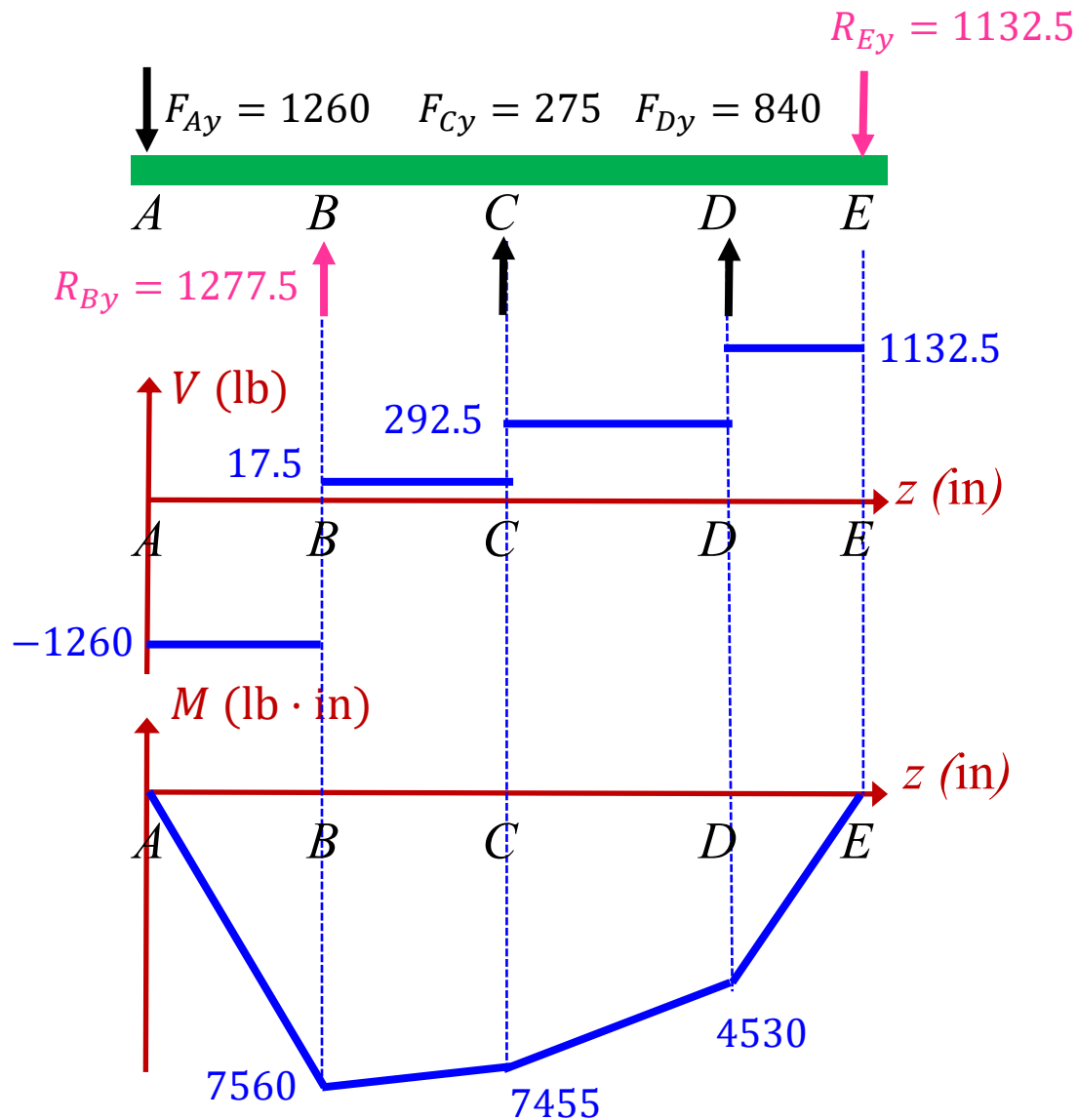
Solutions to Assignment #1, cont'd:

Step 3: Shear force and bending moment diagrams.

xz-plane / horizontal plane



Solutions to Assignment #1, cont'd:



yz-plane / vertical plane

Solutions to Assignment #1, cont'd:

Total bending moment at point *B*:

$$M_B = \sqrt{0^2 + 7560^2} = 7560 \text{ lb} \cdot \text{in}$$

Total bending moment at point *C*:

$$M_C = \sqrt{3175.2^2 + 7455^2} = 8103 \text{ lb} \cdot \text{in}$$

Total bending moment at point *D*:

$$M_D = \sqrt{907.2^2 + 4530^2} = 4620 \text{ lb} \cdot \text{in}$$

Solutions to Assignment #1, cont'd:

Step 4: Design shaft diameter at different places.

The material is SAE 1117 cold-drawn steel. From Appendix 3, we find $s_y = 65$ ksi, $s_u = 80$ ksi

From Figure 5-11(a), we find the endurance limit $s_n = 30$ ksi. Then, the estimated actual endurance limit is

$$s'_n = s_n C_m C_{st} C_R C_S = (30)(1.0)(1.0)(0.81)(0.80) = 19.44 \text{ ksi} = 19440 \text{ psi}$$

For Point C to the left:

$$T_C(\text{left}) = T_{A-C} = 6300 \text{ lb} \cdot \text{in}, \quad M_C = 8103 \text{ lb} \cdot \text{in}$$

Using design equation for shaft under bending and torsion only, $D =$

$$\left[\frac{32N}{\pi} \sqrt{\left(\frac{K_t M}{s'_n} \right)^2 + \frac{3}{4} \left(\frac{T}{s_y} \right)^2} \right]^{1/3}, \text{ with } K_t = 3 \text{ and } N = 2:$$

$$D_C|_{\text{left}} = \left[\frac{32(2)}{\pi} \sqrt{\left(\frac{3(8103)}{19440} \right)^2 + \frac{3}{4} \left(\frac{6300}{65000} \right)^2} \right]^{1/3} = \boxed{2.945 \text{ in}}$$

Solutions to Assignment #1, cont'd:

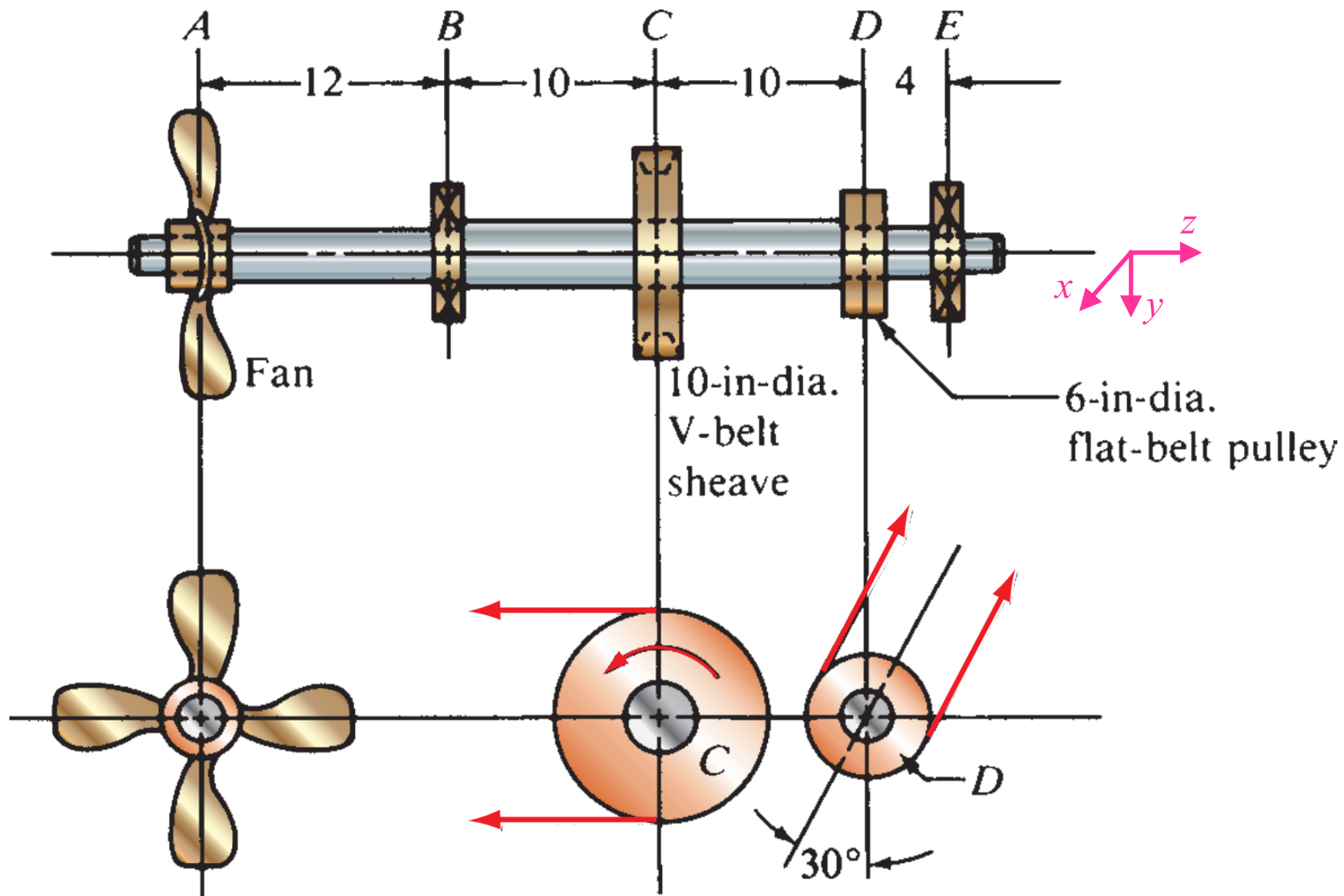
For Point C to the right:

$$T_C(\text{right}) = T_{A-C} = 2520 \text{ lb} \cdot \text{in}, \quad M_C = 8103 \text{ lb} \cdot \text{in}$$

Using design equation for shaft under bending and torsion only,

$$D_C|_{\text{right}} = \left[\frac{32(2)}{\pi} \sqrt{\left(\frac{3(8103)}{19440} \right)^2 + \frac{3}{4} \left(\frac{2520}{65000} \right)^2} \right]^{1/3} = \boxed{2.943 \text{ in}}$$

2. The shaft shown is part of a grain-drying system. At point A is a propeller-type fan that requires 12 hp when rotating at 475 rpm. The fan weighs 34 lb, and its weight should be **included** in the analysis. The flat-belt pulley at point D delivers 3.5 hp to a screw conveyor handling the grain. All power goes into the shaft through the V-belt sheave at point C . Use SAE 1144 cold-drawn steel for the shaft material. (20' in total)
- 1) Determine the torques at A , C , and D . Also determine the torque distribution in A - C , C - D , and D - E region.
 - 2) Plot the shear and bending moment diagrams of the entire shaft in the horizontal and vertical planes.
 - 3) Determine the total bending moment at point C and D .
 - 4) Determine the minimum acceptable diameters for the shaft at point C . Do the design to the left and to the right of point C separately. Take $C_s = 0.85$, $C_R = 0.81$, design factor $N = 3$, stress concentration factor $K_t = 2.0$ (Keyseat) and $K_t = 3.0$ (Ring groove) to the left and to the right of point C , respectively.



Solutions to Assignment #2:

Step 1: Analyze the torque distribution in the shaft.

The propeller-type fan delivers 12 hp at point *A* and the flat-belt pulley at point *D* delivers 3.5 hp. Then, the total power of 15.5 hp goes into the shaft through the V-belt sheave at point *C*.

Torque on fan *A*:

$$T_A = \frac{63000(12 \text{ hp})}{475 \text{ rpm}} = 1592 \text{ lb} \cdot \text{in}$$

Torque on pulley *D*:

$$T_C = \frac{63000(3.5 \text{ hp})}{475 \text{ rpm}} = 464 \text{ lb} \cdot \text{in}$$

Torque on sheave *C*:

$$T_D = \frac{63000(15.5 \text{ hp})}{475 \text{ rpm}} = 2056 \text{ lb} \cdot \text{in}$$

Using Free-Body-Diagram, we get torque distribution in the shaft:

$$\begin{aligned} T_{A-C} &= 1592 \text{ lb} \cdot \text{in} \\ T_{C-D} &= 464 \text{ lb} \cdot \text{in} \\ T_{D-E} &= 0 \text{ lb} \cdot \text{in} \end{aligned}$$

Solutions to Assignment #2, cont'd:

Step 2: Compute forces that act in the vertical and horizontal planes.

(1) Forces on fan *A*:

The forces are due to the weight of the fan.

The components of the bending force at point *A* are:

$$F_{Ax} = 0 \text{ lb}$$

$$F_{Ay} = 34 \text{ lb} \quad (\text{downward or positive } y\text{-direction})$$

(2) Forces on V-belt sheave *C*:

$$\text{Net driving force: } F_N = F_1 - F_2 = \frac{T_C}{\frac{D_C}{2}} = \frac{2056}{\frac{10}{2}} = 411 \text{ lb}$$

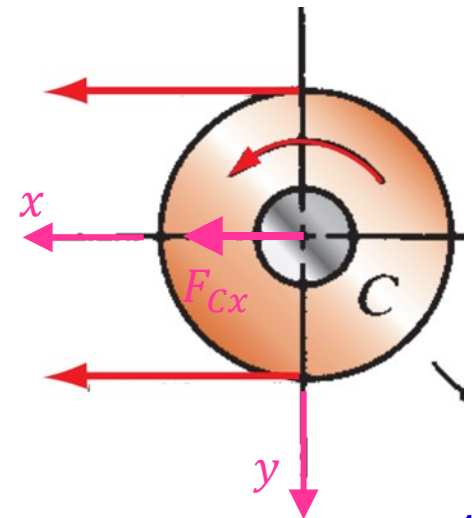
$$\text{Bending force at } C: F_C = 1.5F_N = 1.5(411) = 617 \text{ lb}$$

(Note: for V-belt sheave, a factor of 1.5 should be used)

The components of the bending force at point *C* are:

$$F_{Cx} = 617 \text{ lb} \quad (\text{out-of-plane or positive } x\text{-direction})$$

$$F_{Cy} = 0 \text{ lb}$$



Solutions to Assignment #2, cont'd:

(3) Forces on flat-belt pulley D :

$$\text{Net driving force: } F_N = F_1 - F_2 = \frac{T_D}{\frac{D_D}{2}} = \frac{464}{\frac{6}{2}} = 155 \text{ lb}$$

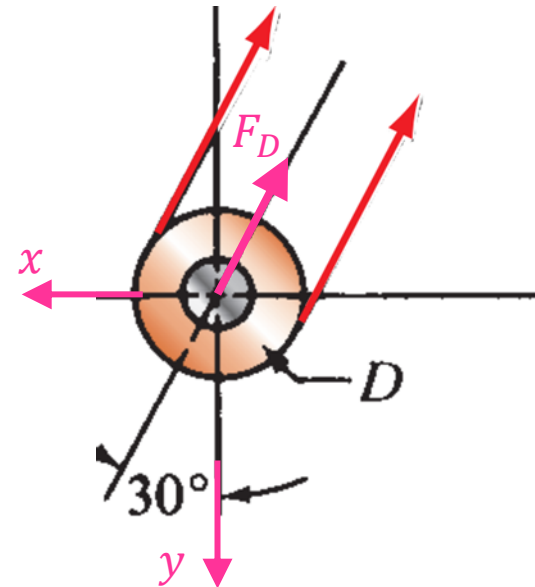
$$\text{Bending force at } D: F_D = 2.0F_N = 2.0(155) = 310 \text{ lb}$$

(Note: for flat-belt pulley, a factor of 2.0 should be used)

The components of the bending force at point D are:

$$F_{Dx} = -F_D \sin 30^\circ = -155 \text{ lb} \quad (\text{negative } x\text{-direction})$$

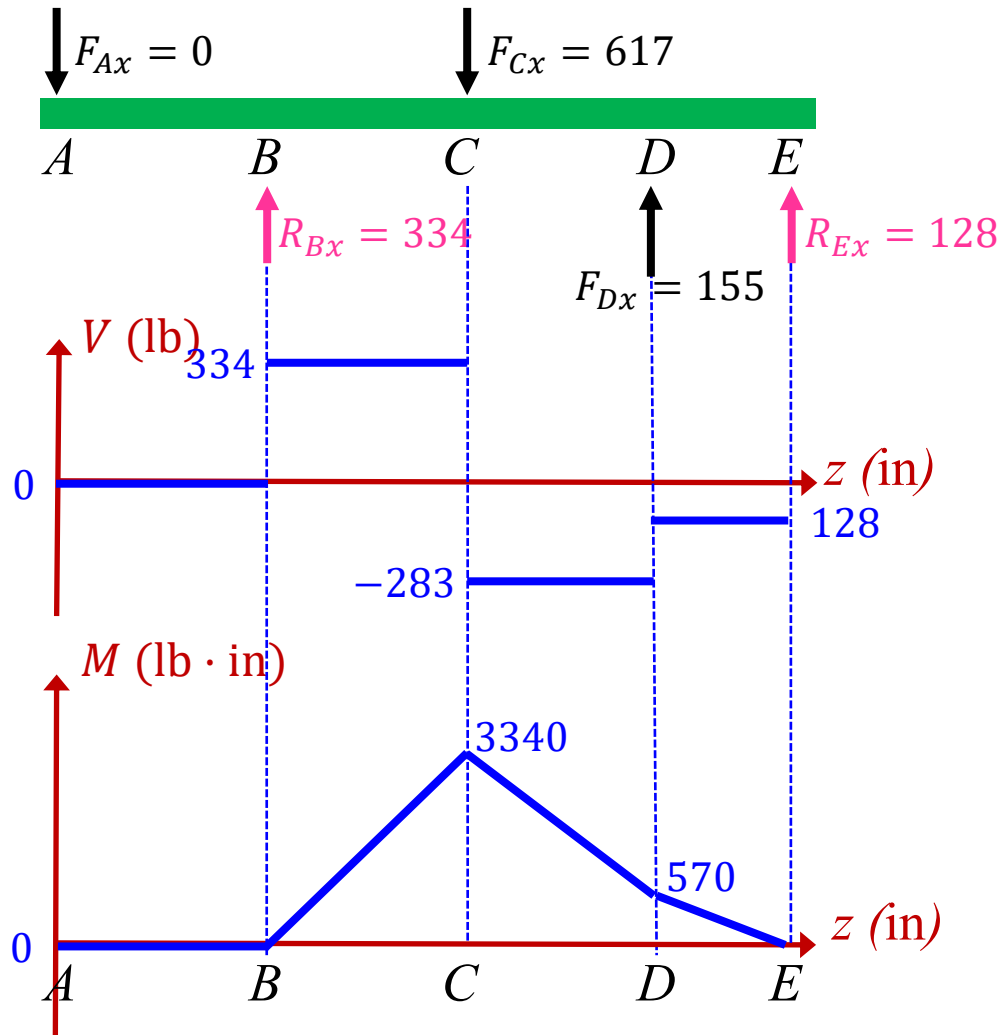
$$F_{Dy} = -F_D \cos 30^\circ = -268 \text{ lb} \quad (\text{upward or negative } y\text{-direction})$$



Solutions to Assignment #2, cont'd:

Step 3: Shear force and bending moment diagrams.

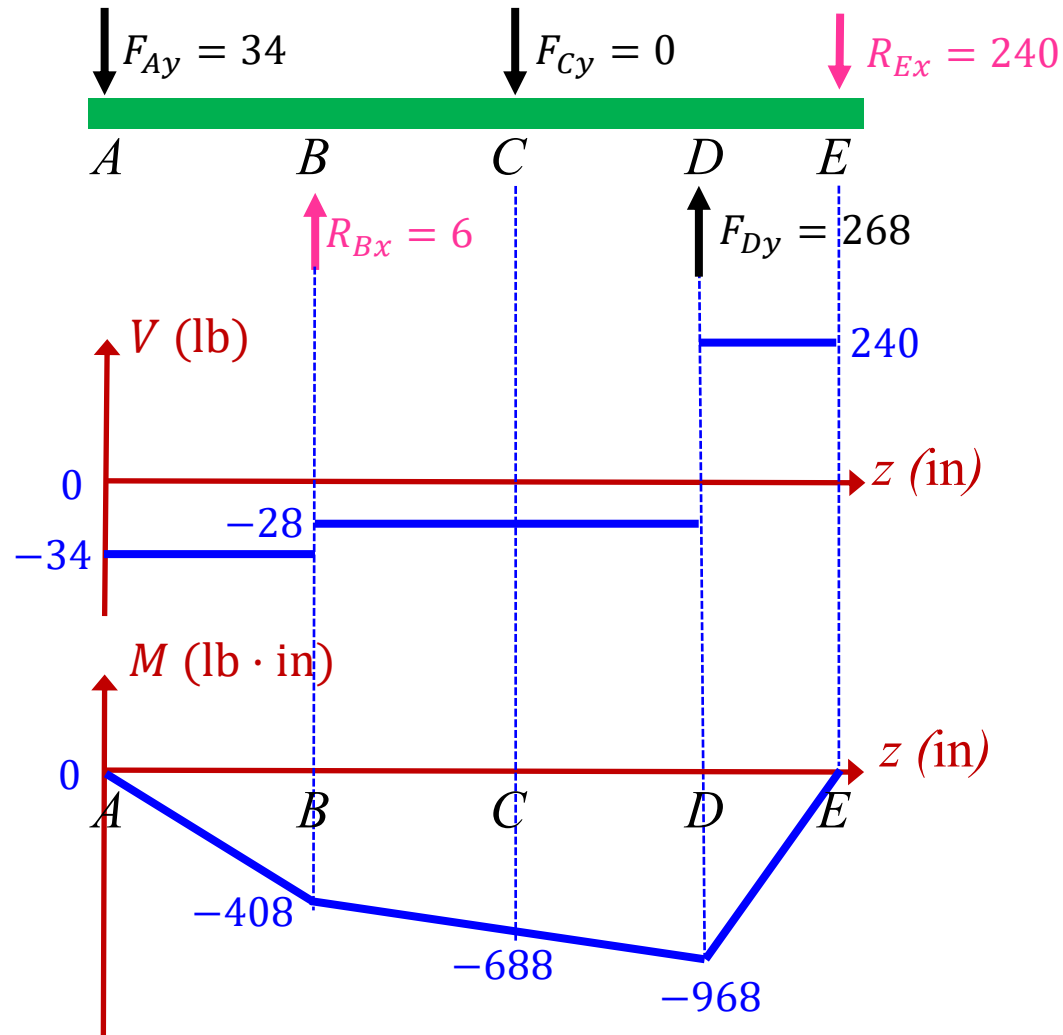
xz-plane / horizontal plane



Solutions to Assignment #2, cont'd:

Step 3: Shear force and bending moment diagrams.

yz-plane / vertical plane



Solutions to Assignment #2, cont'd:

Total bending moment at point C :

$$M_C = \sqrt{3340^2 + 688^2} = 3410 \text{ lb} \cdot \text{in}$$

Total bending moment at point D :

$$M_D = \sqrt{570^2 + 968^2} = 1123 \text{ lb} \cdot \text{in}$$

Solutions to Assignment #2, cont'd:

Step 4: Design shaft diameter at different places.

The material is SAE 1144 cold-drawn steel. From Appendix 3, we find $s_y = 90$ ksi, $s_u = 100$ ksi

From Figure 5-11(a), we find the endurance limit $s_n = 38$ ksi. Then, the estimated actual endurance limit is

$$s'_n = s_n C_m C_{st} C_R C_S = (38)(1.0)(1.0)(0.85)(0.81) = 26.163 \text{ ksi} = 26163 \text{ psi}$$

For Point C to the left (Keyseat):

$$T_C(\text{left}) = T_{A-C} = 1592 \text{ lb} \cdot \text{in}, \quad M_C = 3410 \text{ lb} \cdot \text{in}$$

Using design equation for shaft under bending and torsion only, $D =$

$$\left[\frac{32N}{\pi} \sqrt{\left(\frac{K_t M}{s'_n} \right)^2 + \frac{3}{4} \left(\frac{T}{s_y} \right)^2} \right]^{1/3}, \text{ with } K_t = 2.0 \text{ and } N = 3:$$

$$D_C|_{\text{left}} = \left[\frac{32(3)}{\pi} \sqrt{\left(\frac{2.0(3410)}{26163} \right)^2 + \frac{3}{4} \left(\frac{1592}{90000} \right)^2} \right]^{1/3} = \boxed{1.998 \text{ in}}$$

Solutions to Assignment #2, cont'd:

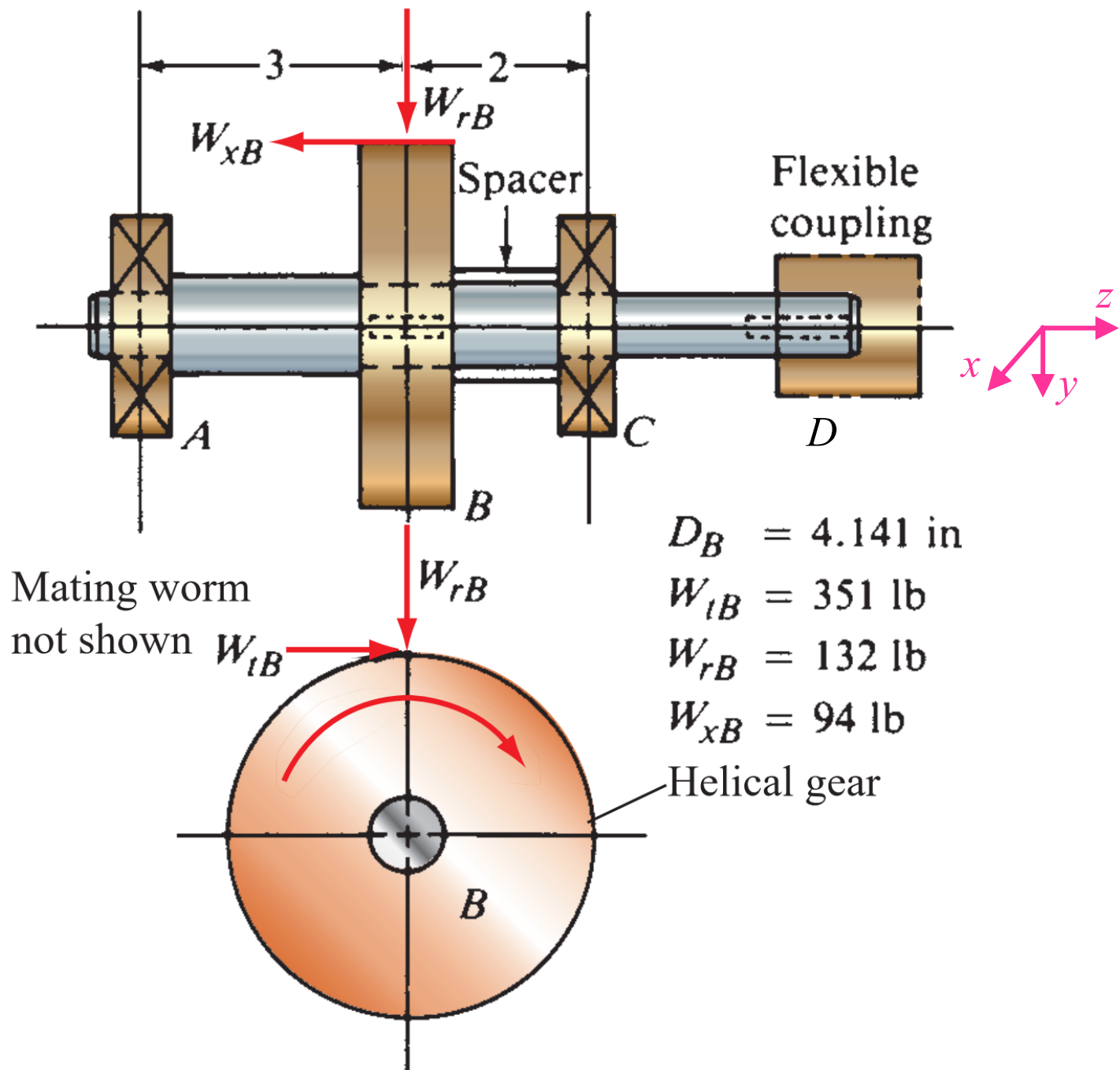
For Point C to the right (Ring groove):

$$T_C(\text{right}) = T_{C-D} = 464 \text{ lb} \cdot \text{in}, \quad M_C = 3410 \text{ lb} \cdot \text{in}$$

Using design equation for shaft under bending and torsion only,

$$D_C|_{\text{right}} = \left[\frac{32(3)}{\pi} \sqrt{\left(\frac{3.0(3410)}{26163} \right)^2 + \frac{3}{4} \left(\frac{464}{90000} \right)^2} \right]^{1/3} = \boxed{2.286 \text{ in}}$$

3. A helical gear mounted on a shaft that rotates at 650 rpm while transmitting 7.5 hp. The tangential, radial, and axial forces on the gear are shown in the figure. The pitch diameter of the gear is 4.141 in. The power is delivered from the shaft through a flexible coupling at its right end. A spacer is used to position the gear relative to bearing *C*. The thrust load is taken at bearing *A*. Use SAE 1020 cold-drawn steel for the shaft material. (20' in total)
- 1) Determine the torque at *B*. Also determine the torque distribution in *A-B*, *B-C*, and *C-D* region.
 - 2) Plot the shear and bending moment diagrams for the *A-B-C* region of the shaft in the horizontal and vertical planes.
 - 3) Determine the total bending moment at point *B* (for both to the left and to the right).
 - 4) Determine the minimum acceptable diameters for the shaft at point *B*. Do the design to the left and to the right of point *B* separately. Take $C_s = 0.90$, $C_R = 0.81$, design factor $N = 3$, stress concentration factor $K_t = 2.0$.



Solutions to Assignment #3:

Step 1: Analyze the torque distribution in the shaft.

The gear delivers 7.5 hp at point B . Then, the torque at the gear B :

$$T_B = \frac{63000(7.5 \text{ hp})}{650 \text{ rpm}} = \boxed{727 \text{ lb} \cdot \text{in}}$$

Since the power is delivered to the flexible coupling at the right end, using Free-Body-Diagram, we get torque distribution in the shaft:

$$\begin{aligned} T_{A-B} &= 0 \text{ lb} \cdot \text{in} \\ T_{B-C} &= T_{C-D} = 727 \text{ lb} \cdot \text{in} \end{aligned}$$

Solutions to Assignment #2, cont'd:

Step 2: Compute forces that act in the vertical and horizontal planes.

Forces on gear B :

With the given tangential, radial, and axial forces on the gear as shown in the figure, we assign them as the components of the bending forces:

$$F_{Bx} = -W_{tB} = -351 \text{ lb} \quad (\text{inside of plane or negative } x\text{-direction})$$

$$F_{By} = W_{rB} = 132 \text{ lb} \quad (\text{downward or positive } y\text{-direction})$$

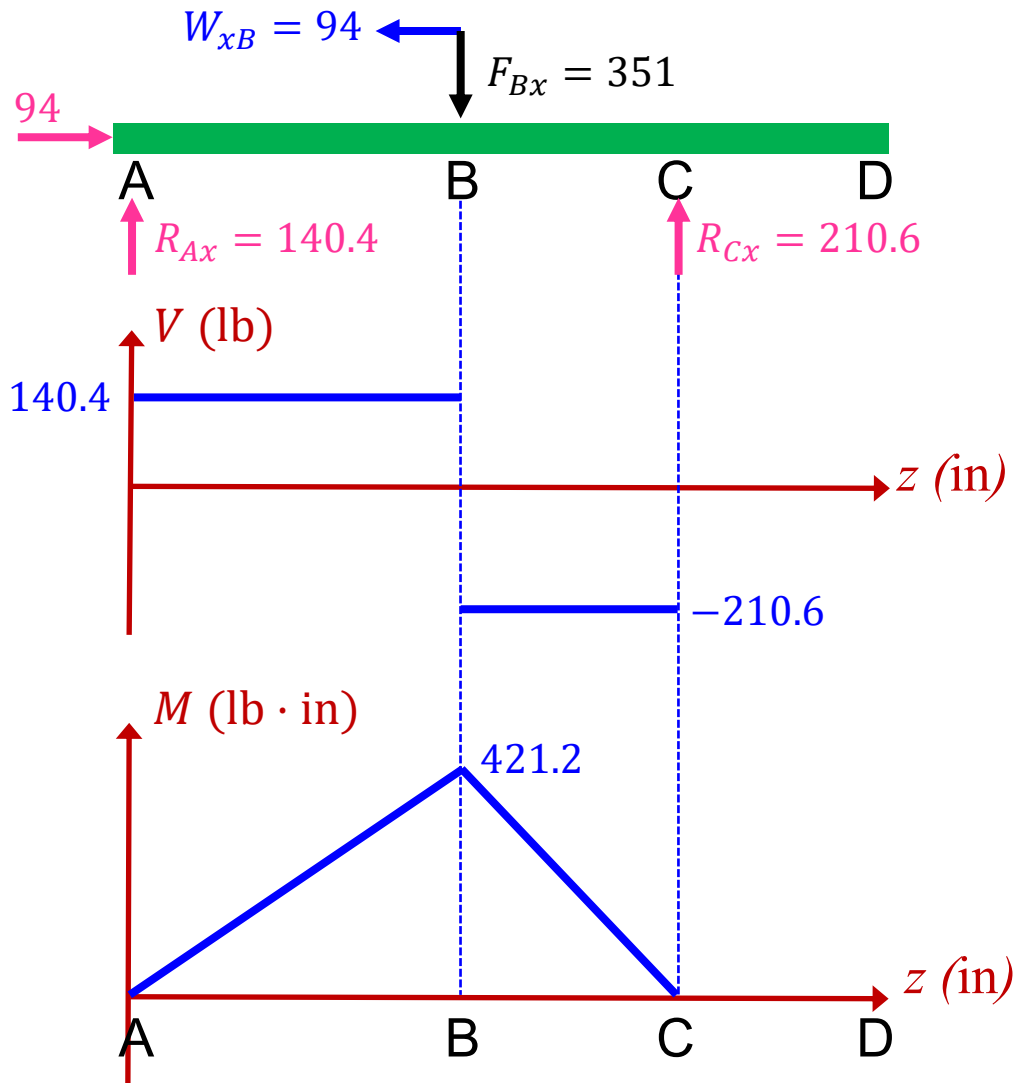
Additional bending moment at B due to the off-center axial force (W_{xB}):

$$M_B = W_{xB} \times \frac{D_B}{2} = (94 \text{ lb}) \left(\frac{4.141}{2} \text{ in} \right) = 194.6 \text{ lb} \cdot \text{in}$$

Solutions to Assignment #3, cont'd:

Step 3: Shear force and bending moment diagrams.

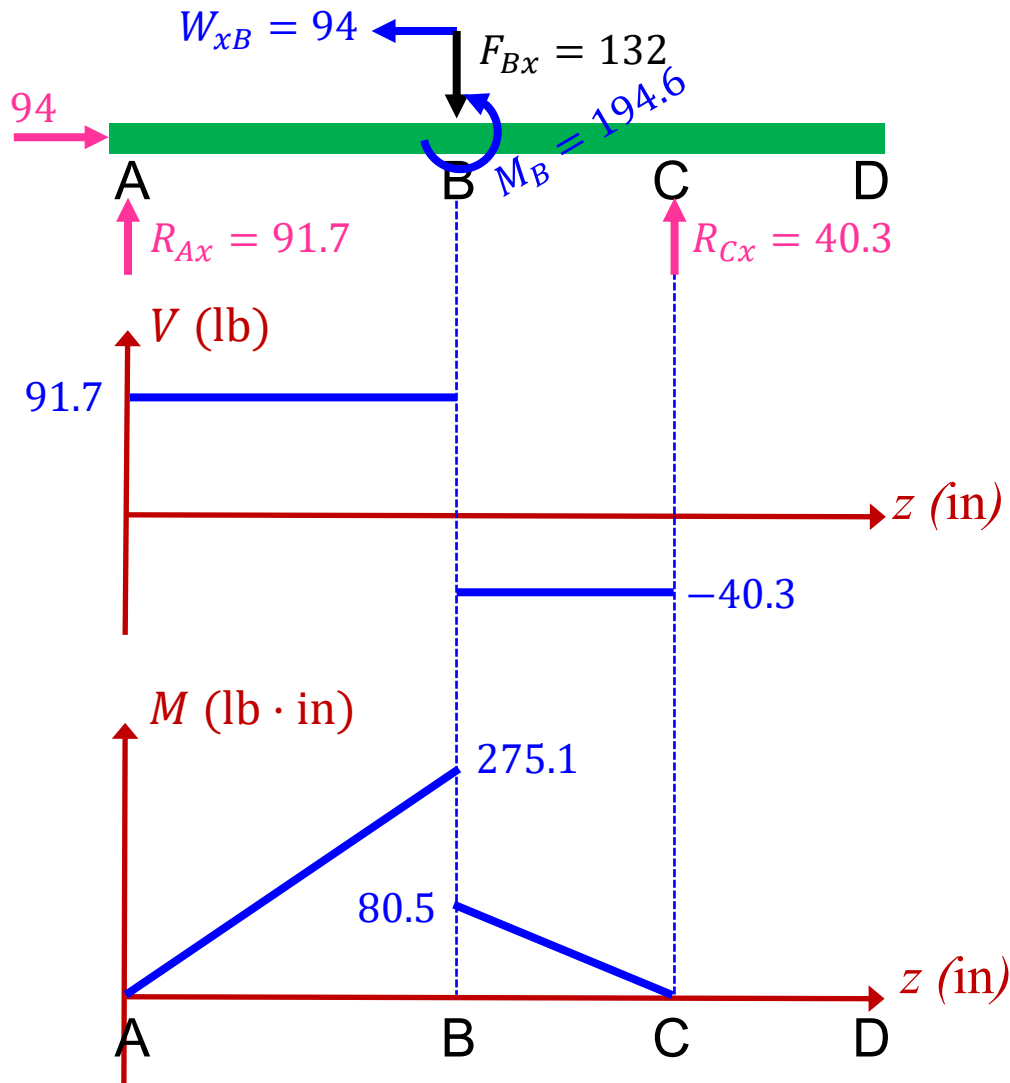
xz-plane / horizontal plane



Solutions to Assignment #3, cont'd:

Step 3: Shear force and bending moment diagrams.

yz-plane / vertical plane



Solutions to Assignment #3, cont'd:

Total bending moment at point B to the left:

$$M_B|_{\text{left}} = \sqrt{421.2^2 + 275.1^2} = 503 \text{ lb} \cdot \text{in}$$

Total bending moment at point D :

$$M_B|_{\text{right}} = \sqrt{421.2^2 + 80.5^2} = 429 \text{ lb} \cdot \text{in}$$

Solutions to Assignment #3, cont'd:

Step 4: Design shaft diameter at point B .

The material is SAE 1020 cold-drawn steel. From Appendix 3, we find $s_y = 51$ ksi, $s_u = 61$ ksi

From Figure 5-11(a), we find the endurance limit $s_n = 22$ ksi. Then, the estimated actual endurance limit is

$$s'_n = s_n C_m C_{st} C_R C_S = (22)(1.0)(1.0)(0.81)(0.90) = 16.038 \text{ ksi} = 16038 \text{ psi}$$

For point B to the left (Keyseat):

$$T_B(\text{left}) = T_{A-B} = 0 \text{ lb} \cdot \text{in}, \quad M_B|_{\text{left}} = 503 \text{ lb} \cdot \text{in}$$

Using design equation for shaft under bending and torsion only, with $K_t = 2.0$ and $N = 3$:

$$D_B|_{\text{left}} = \left[\frac{32(3)}{\pi} \sqrt{\left(\frac{2.0(503)}{16038} \right)^2 + \frac{3}{4} \left(\frac{0}{51000} \right)^2} \right]^{1/3} = \boxed{1.242 \text{ in}}$$

Solutions to Assignment #3, cont'd:

For point B to the right (Ring groove):

$$T_B(\text{right}) = T_{B-C} = 727 \text{ lb} \cdot \text{in}, \quad M_B|_{\text{right}} = 429 \text{ lb} \cdot \text{in}$$

Using design equation for shaft under bending and torsion only,

$$D_B|_{\text{right}} = \left[\frac{32(3)}{\pi} \sqrt{\left(\frac{2.0(429)}{16038} \right)^2 + \frac{3}{4} \left(\frac{727}{51000} \right)^2} \right]^{1/3} = \boxed{1.188 \text{ in}}$$

4. A helical compression spring with free length of 68 mm needs to carry a load of 500 N with a deflection of 25 mm. The spring index can be taken as 8. Assume the following values for spring material: the resultant maximum shear stress $\tau_{\max} = 350$ MPa and shear modulus of elasticity $G = 84$ GPa. Design this spring by specifying following: (**5'** in total)

- 1) wire diameter (D_w)
- 2) mean diameter (D_m)
- 3) number of active coils (N_a)
- 4) total number of coils (N) assuming squared and ground ends
- 5) solid length of the spring (L_s)
- 6) pitch of the spring (p)

Solutions to Assignment #4:

Stiffness of the spring can be determined by:

$$k = \frac{\Delta F}{\Delta L} = \frac{500 \text{ N}}{25 \text{ mm}} = 20 \frac{\text{N}}{\text{mm}} = 20000 \text{ N/m}$$

Spring index is given by $C = \frac{D_m}{D_w}$. Since $C = 8$, thus $\frac{D_m}{D_w} = 8$, i.e.,

$$D_m = 8D_w \quad (1)$$

For load of 500 N, the resultant maximum shear stress is given by

$$\tau_{\max} = \frac{8KFD_m}{\pi D_w^3} = \frac{8KFC}{\pi D_w^2}, \text{ with Wahl Factor } K = \frac{4C-1}{4C-4} + \frac{0.615}{C} = \frac{4(8)-1}{4(8)-4} + \frac{0.615}{8} = 1.184$$

i.e.,

$$350 \times 10^6 = \frac{8(1.184)(500)(8)}{\pi D_w^2} \quad (2)$$

Solving Eqs. (1) & (2) we get

$\begin{aligned} D_w &= 0.00587 \text{ m} = 5.87 \text{ mm} \\ D_m &= 46.96 \text{ mm} \end{aligned}$

Solutions to Assignment #4, cont'd:

Linear Deflection is defined as

$$f = \frac{8FD_m^3 N_a}{GD_w^4} = \frac{8FC^3 N_a}{GD_w}$$

Therefore, the active number of coils can be calculated as

$$N_a = \frac{fGD_w}{8FC^3} = \frac{(0.025)(84 \times 10^9)(0.00587)}{8(500)(8^3)} = 6.019 = \boxed{7}$$

For squared and ground ends, the total number of coils is

$$N = N_a + 2 = 7 + 2 = \boxed{9}$$

Solid length of the spring is given by

$$L_s = ND_w = 9(5.87) = \boxed{52.83 \text{ mm}}$$

Finally, according to the relationship between free length and pitch, $L_f = pN_a + 2D_w$, we can compute pitch as

$$p = \frac{L_f - 2D_w}{N_a} = \frac{68 - 2(5.87)}{7} = \boxed{8.04 \text{ mm}}$$

5. A support bar for a machine component is suspended to soften applied loads. The mean diameter of the spring must be 3.00 in. The installation length of the spring is 3.00 in and corresponding force is 180 lb. During operation, the load on the spring is 220 lb with the position of the rod is to move 0.50 in. Several million cycles of load application are expected. Use ASTM A229 steel wire. The type of end condition is squared and ground ends. Design a **standard** helical compression spring by specifying following: (20' in total)

- 1) wire diameter, mean diameter, spring index, Wahl factor
- 2) OD, ID, free length
- 3) solid length and corresponding force
- 4) number of active coils and total number of coils
- 5) determine actual design stress
- 6) check the maximum shear stress at the operating load and at the solid length condition

Solutions to Assignment #5:

Step 1: Since the spring material is ASTM A229 steel, from Table 18-4, we get the shear modulus of elasticity $G = 11.2 \times 10^6$ psi.

Step 2:

Operating force $F_o = 220$ lb, installation force $F_i = 180$ lb, $\Delta F = 220 - 180 = 40$ lb, the corresponding deflection of the spring is $\Delta L = 0.50$ in

Stiffness of the spring can be determined by:

$$k = \frac{\Delta F}{\Delta L} = \frac{40 \text{ lb}}{0.5 \text{ in}} = 80 \text{ lb/in}$$

Step 3: The free length corresponds to the length of the spring when the external force is zero, i.e.,

$$k = \frac{F_i - F_f}{L_f - L_i} = \frac{F_i - 0}{L_f - L_i} = \frac{F_i}{L_f - L_i}$$

Thus, we get the free length:

$$L_f = \frac{F_i}{k} + L_i = \frac{180}{80} + 3.00 = \boxed{5.25 \text{ in}}$$

Solutions to Assignment #5, cont'd:

Step 3: Since several million cycles of load application are expected, the loading belongs to the **severe** service.

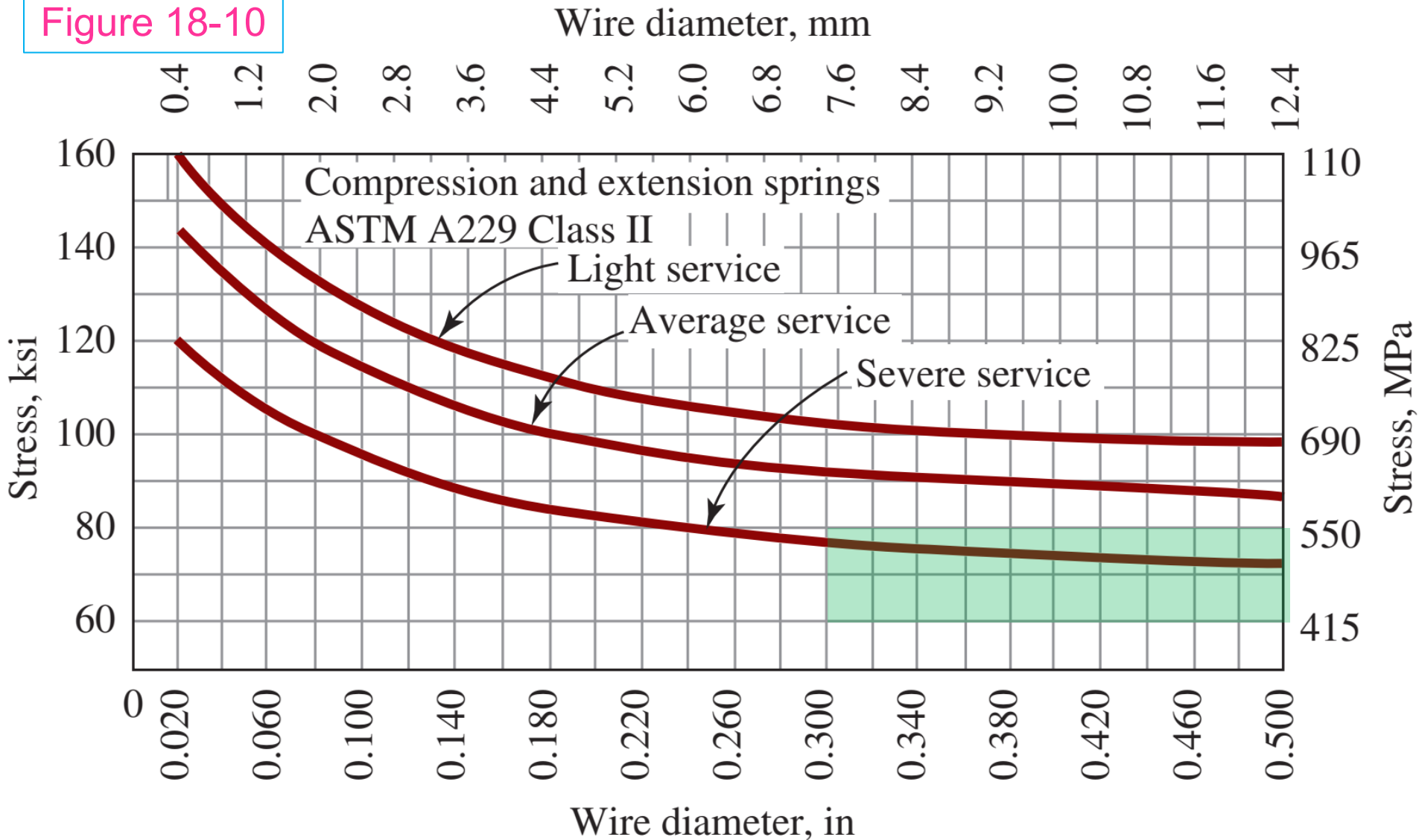
Step 4: Try an initial design stress.

For the ASTM A229 steel, as shown in Figure 18–10, the **nominal** design stress would be in the range of 70 000 to 80 000 psi, using the rough estimate of $D_w = D_m/C = 0.3 - 0.6$, corresponding spring index $C = 5 - 10$, (see next page). We can take a medium value of **75 000 psi** for design stress.

The design stress should be checked later.

Solutions to Assignment #5, cont'd:

Figure 18-10



Solutions to Assignment #5, cont'd:

Step 5: Compute the trial wire diameter (D_w).

As shown in the In-Class Example in Ch9.2, the Wahl factor changes from 1.15 to 1.30 for a good estimate of spring index $C = 5 - 10$. Thus, we can take a medium value of 1.20 for the Wahl factor. Then, we can solve the wire diameter as

$$D_w^3 = \frac{8KFD_m}{\pi\tau_{\max}} = \frac{8(1.20)(220)(3.00)}{\pi(75000)} = 2.6891 \times 10^{-2}$$

$$D_w = 0.2996 \text{ in}$$

The next larger **standard** wire size is 0.3065 in, U.S. steel wire gage no. 0, see Table 18-2.

Then, we select $D_w =$ 0.3065 in

From Figure 18-10, using $D_w = 0.3065$ in, we find the actual design stress is

$\tau_d = 78$ ksi for severe service, and maximum allowable shear stress is 102 ksi for light service.

Solutions to Assignment #5, cont'd:

Step 6: Recompute the actual values of spring index and Wahl factor.

$$C = \frac{D_m}{D_w} = \frac{3.00}{0.3065} = \boxed{9.79}$$

$$K = \frac{4C - 1}{4C - 4} + \frac{0.615}{C} = \frac{4(9.79) - 1}{4(9.79) - 4} + \frac{0.615}{9.79} = \boxed{1.15}$$

Both spring index and Wahl factor are in the estimated range. **OK**

Solutions to Assignment #5, cont'd:

Step 7: Recompute and check the actual shear stress under operation force:

$$\tau_{\max}|_o = \frac{8KF_oD_m}{\pi D_w^3} = \frac{8(1.15)(220)(3.00)}{\pi(0.3065)^3} = 67126 \text{ psi} < 78000 \text{ psi}$$

The shear stress is safe.

Step 8: Determine the number of active coils from linear deflection, $f = \frac{8FD_m^3N_a}{GD_w^4} = \frac{8FC^3N_a}{GD_w}$ (note spring rate $k = \frac{F}{f} = 80 \text{ lb/in}$ is already obtained),

$$N_a = \frac{GD_w}{8kC^3} = \frac{(11.2 \times 10^6)(0.3065)}{8(80)(9.79)^3} = 5.72 = \boxed{6}$$

For squared and ground ends, the total number of coils is

$$N = N_a + 2 = 6 + 2 = \boxed{8}$$

Solutions to Assignment #5, cont'd:

Step 9: Compute the solid length:

$$L_s = N \times D_w = (8)(0.3065) = \boxed{2.452 \text{ in}}$$

The corresponding force at the solid length can be determined by the product of the spring rate times the deflection to the solid length ($L_f - L_s$):

$$F_s = k(L_f - L_s) = 80(5.25 - 2.452) = \boxed{223.84 \text{ lb}}$$

According to the maximum shear stress formula, $\tau_{\max} = \frac{8KFD_m}{\pi D_w^3}$, we know that the shear stress is proportional to the force, i.e., the shear stress at the solid length can be computed as

$$\tau_s = \tau_o \left(\frac{F_s}{F_o} \right) = (67126) \frac{223.84}{220} = 68298 \text{ psi} < 78000 \text{ psi}$$

The spring is safe when compressed to solid length.

Solutions to Assignment #5, cont'd:

Step 10: Determine other geometry of the spring:

$$\text{Outer diameter: } OD = D_m + D_w = 3.00 + 0.3065 = 3.3065 \text{ in}$$

$$\text{Inner diameter: } ID = D_m - D_w = 3.00 - 0.3065 = 2.6935 \text{ in}$$

6. A pair of spur gears with 25° , full-depth, involute teeth transmits 50 hp. The pinion is mounted on the shaft of an electric motor operating at 1150 rpm. The pinion has 18 teeth and a diametral pitch of 5. The gear has 68 teeth. Compute the following: (**5'** in total)

- 1) The rotational speed of the gear
- 2) The velocity ratio and the gear ratio for the gear pair
- 3) The pitch diameter of the pinion and the gear
- 4) The center distance between the shafts carrying the pinion and the gear
- 5) The pitch line speed for both the pinion and the gear
- 6) The torque on the pinion shaft and on the gear shaft
- 7) The tangential force acting on the teeth of each gear
- 8) The radial force acting on the teeth of each gear
- 9) The normal force acting on the teeth of each gear

Solutions to Assignment #6:

Given: $P = 50$ hp, $n_P = 1150$ rpm, $N_P = 18$, $N_G = 68$, $P_d = 5$ in⁻¹, $\phi = 25^\circ$

1) Since $\frac{n_P}{n_G} = \frac{N_G}{N_P}$ (speed reduction equation), therefore

$$n_G = \frac{n_P N_P}{N_G} = \frac{1150(18)}{68} = \boxed{304.4 \text{ rpm}}$$

$$2) \text{ Velocity Ratio } VR = \frac{n_P}{n_G} = \frac{N_G}{N_P} = \frac{68}{18} = 3.778$$

$$\text{Gear Ratio } m_G = \frac{N_G}{N_P} = \boxed{3.778}$$

3) According to Diametral Pitch definition, $P_d = \frac{N_G}{D_G} = \frac{N_P}{D_P}$, we get the pitch diameter of the gear $D_G = \frac{N_G}{P_d} = \frac{68}{5 \text{ in}^{-1}} = \boxed{13.600 \text{ in}}$

$$\text{Pitch diameter of the pinion } D_P = \frac{N_P}{P_d} = \frac{18}{5 \text{ in}^{-1}} = \boxed{3.600 \text{ in}}$$

Solutions to Assignment #6, cont'd:

4) The center distance between the shafts carrying the pinion and the gear

$$C = \frac{D_P + D_G}{2} = \frac{13.600 + 3.600}{2} = \boxed{8.600 \text{ in}}$$

5) The pitch line speed for both the pinion and the gear

$$v_t = \left(\frac{D}{2}\right) \omega = \frac{\pi D n}{12} = \frac{\pi D_P n_P}{12} = \frac{\pi (3.600)(1150)}{12} = \boxed{1084 \text{ ft/min}}$$

6) The torque on the pinion shaft and on the gear shaft

$$T_P = 63\,000 \frac{P}{n_P} = 63\,000 \frac{50}{1150} = \boxed{2739 \text{ lb} \cdot \text{in}}$$

$$T_G = 63\,000 \frac{P}{n_G} = 63\,000 \frac{50}{304.4} = \boxed{10348 \text{ lb} \cdot \text{in}}$$

Solutions to Assignment #6, cont'd:

7) The tangential force acting on the teeth of each gear

$$W_t = \frac{T_P}{\frac{D_P}{2}} = \frac{2739}{3.600/2} = \boxed{1522 \text{ lb}}$$

Or

$$W_t = \frac{33\,000\,P}{v_t} = \frac{33\,000\,(50)}{1084} = 1522 \text{ lb}$$

8) The radial force acting on the teeth of each gear

$$W_r = W_t \tan \phi = (1522) \tan 25^\circ = \boxed{710 \text{ lb}}$$

9) The normal force acting on the teeth of each gear

$$W_n = W_t / \cos \phi = 1522 / \cos 25^\circ = \boxed{1679 \text{ lb}}$$

Or

$$W_n = \sqrt{W_t^2 + W_r^2} = \sqrt{1522^2 + 710^2} = 1679 \text{ lb}$$

7. Devise a gear train using all external gears on parallel shafts. Use 20° full-depth involute teeth and no less than 16 teeth and no more than 150 teeth in any gear. The input speed is 1800 rpm and the output speed must be **exactly** 2.0 rpm. Ensure that there is no interference. Sketch the layout for your design. (**10'** in total)

Solutions to Assignment #7:

Given: $n_{\text{input}} = 1800 \text{ rpm}$, $n_{\text{output}} = 2 \text{ rpm}$, exact ratio required.

First, we compute the nominal train value (TV):

$$TV = (\text{input speed})/(\text{output speed}) = 1800/2 = 900$$

Factors are: $2 \times 2 \times 5 \times 5 \times 3 \times 3$

For 20° full-depth involute teeth, $N_{\text{min}} = 16$, $N_{\text{max}} = 150$

Nominal maximum velocity ratio per pair:

$$VR_{\text{max}} = \frac{N_{\text{max}}}{N_{\text{min}}} = \frac{150}{16} = 9.375$$

Single gear pair: $VR = 9.375$

(Too small)

Two pairs: $VR = (9.375)^2 = 87.89$

(Too small)

Three pairs: $VR = (9.375)^3 = 823.97$

(Still smaller than 900)

Four pairs: $VR = (9.375)^4 = 7724$

(fulfill requirement, four pairs required)

Solutions to Assignment #7, cont'd:

Recombine VR factors: $VR_1 = 6$, $VR_2 = 6$, $VR_3 = 5$, $VR_4 = 5$

$N_A = 16$	$N_C = 16$	$N_E = 16$	$N_G = 16$
$N_B = 96$	$N_D = 96$	$N_F = 80$	$N_H = 80$
$\left \begin{array}{c} 16 \\ 96 \end{array} \right 6$	$\left \begin{array}{c} 16 \\ 96 \end{array} \right 6$	$\left \begin{array}{c} 16 \\ 80 \end{array} \right 5$	$\left \begin{array}{c} 16 \\ 80 \end{array} \right 5$

Check final $TV = \frac{96}{16} \times \frac{96}{16} \times \frac{80}{16} \times \frac{80}{16} = 900$ **Exact, OK!**

The schematic gear train is shown below.

